

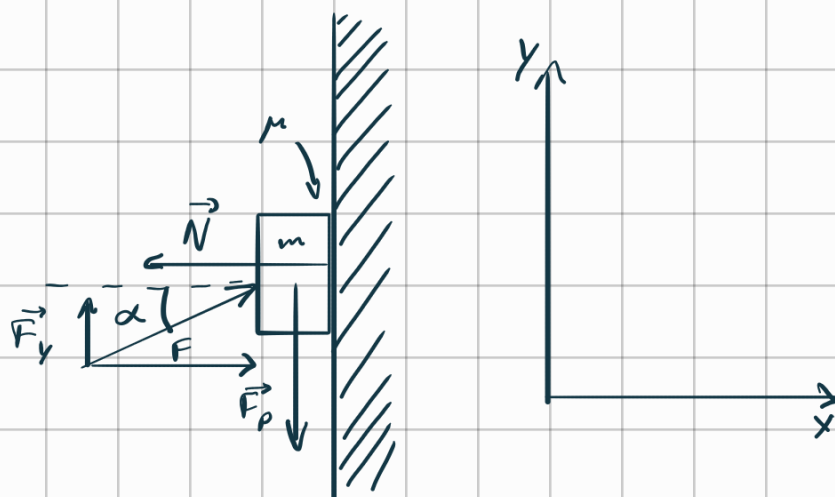
ES. 43

$$m = 2 \text{ kg}$$

$$\alpha = 30^\circ$$

$$F = 20 \text{ N}$$

$$\mu = 1$$



1) trovare tutte le varie forze

$$\text{ASSE } x: \vec{F}_x, \vec{N} \mapsto \vec{F}_x + \vec{N} = 0$$

$$\text{ASSE } y: \vec{F}_y, \vec{F}_a, \vec{F}_p \rightarrow \vec{F}_y + \vec{F}_a + \vec{F}_p = 0$$

2) Supponiamo di trascurare l'attrito e vediamo se vince $\vec{F}_y$ o $\vec{F}_p$

$$F_y - F_p = F \sin \alpha - mg = 20 \text{ N} \cdot \sin 30 - 2 \text{ kg} \cdot 9,81 \text{ N} =$$

$$\downarrow = -9,62 \text{ N} < 0 \rightarrow F_p > F_y$$

neessitiamo quindi di una forza di attrito che si opponga alla forza peso per avere l'equilibrio statico

3) calcolo la forza di attrito

$$\begin{aligned} \vec{F}_a &= \mu_s \cdot \vec{N} \rightarrow F \cos \alpha - N = 0 \rightarrow N = F \cos \alpha \rightarrow \\ &\rightarrow F_a = \mu_s \cdot F \cos \alpha \approx 17,3 \text{ N} > 9,62 \text{ N} \end{aligned}$$

4) calcolo il valore massimo che la massa può assumere

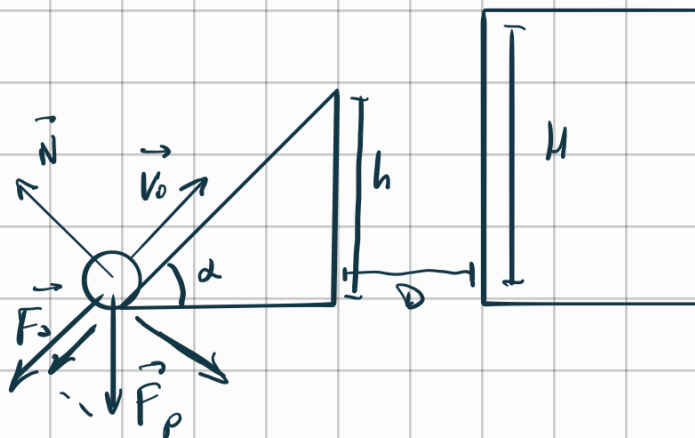
$$\alpha = 45^\circ$$

$$h = 1 \text{ m}$$

$$\mu = 0,5$$

$$D = 80 \text{ cm}$$

$$H = 1,4 \text{ m}$$



$$V_1 = ?$$

1) bilancio delle forze sull'asse x e y ($x \parallel$ piano)

$$x: \vec{F}_p \parallel + \vec{F}_A = m \cdot a$$

$$\hookrightarrow m g \sin \alpha + \vec{F}_A = -m \cdot a \rightarrow \cancel{m} g \sin \alpha + \cancel{m} g \cos \alpha = -\cancel{m} a$$

$$a = -(g \sin \alpha + g \cos \alpha) = -10 \text{ m/s}^2 *$$

$$y: N = \vec{F}_{p\perp} = m g \cos \alpha$$

$$V_1^2 = V_0^2 + 2 a \cdot s$$

$$\begin{cases} V = V_0 + a \cdot t \\ s = s_0 + V_0 \cdot t + \frac{1}{2} a t^2 \end{cases} \rightarrow s = V_0 \left(\frac{V - V_0}{a} \right) + \frac{1}{2} a \left(\frac{V - V_0}{a} \right)^2$$

$$s = \frac{h}{\sin \alpha} = \sqrt{2} h$$

$$V_1^2 = V_0^2 - 2 g \frac{1+\mu}{\sqrt{2}} \cdot \sqrt{2} h = V_0^2 - 2 g (1+\mu) h$$

$$V_1^2 = V_0^2 - 3 g h$$

studio il moto di rot.

$$V_{1x} = V_{1y} = \frac{V_1}{\sqrt{2}}$$

$$\begin{cases} x(t) = x/0 + v_{ix} \cdot t^* \rightarrow t^* = \frac{D}{v_{Ax}} = \frac{D\sqrt{2}}{v_i} \\ y(t) = h + v_{iy} t^* - \frac{1}{2} g t^{*2} \end{cases}$$

$$y\left(\frac{D\sqrt{2}}{v_i}\right) \geq H$$

$$h + D - \frac{g D^2}{v_i^2} \geq H \quad \downarrow$$

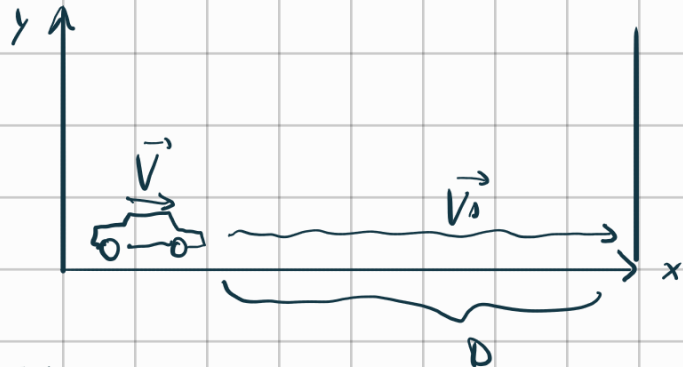
$$v_i^2 \geq \frac{g D^2}{h + D - H}$$

$$\frac{g D^2}{h + D - H} = v_0^2 - 3gh \approx 6,7 \text{ m/s}$$

$$D = 1 \text{ km}$$

$$\Delta t = 6 \text{ s}$$

$$v_s = 360 \text{ m/s}$$



$$\vec{x}_m(t) = \vec{x}_0 + \vec{V} t \rightarrow x(t) = v \cdot t$$

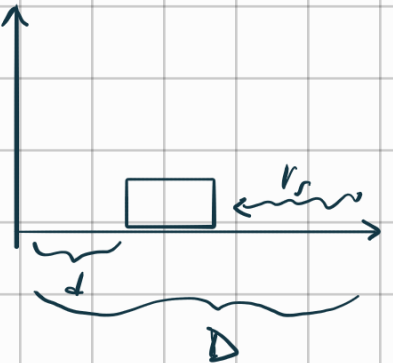
$$\vec{x}_s(t) = v_s \cdot t$$

$$t^* = \frac{D}{v_s}$$

$$x'(t) = -v_s t$$

$$t' = \frac{D - d}{v_s}$$

$$x'(t) = D - v_s t - d$$



$$t^* + t' = \Delta t = \frac{D}{v_s} + \frac{D - d}{v_s} = \frac{2D - d}{v_s}$$

$$d = v \cdot \Delta t$$

$$d = 2D - V_s \Delta t$$

$$2D - V_s \Delta t = V \Delta t$$

$$\rightarrow V = \frac{2D - v_s \Delta t}{\Delta t} =$$